FAQs

1. Define data, data item and entity.

Ans:

- Data: A collection of values or facts.
- Data item: A single unit of data.
- Entity: An object that has attributes and values.

2. Define data structure.

Ans: A data structure is a way of organizing and storing data so that it can be accessed and modified efficiently.

3. What are the types of data structures? Give examples.

Ans: Data structures are classified into:

- Primitive: int, char, float.
- Non-primitive:
    - Linear: Arrays, linked lists, stacks, queues.
    - Non-linear: Trees, graphs.

4. What is a nested structure?

Ans: A nested structure is a structure defined inside another structure in C.

5. What are unions?

Ans: A union is a user-defined data type in C that allows different data types to share the same memory location.

6. Compare structures with unions.

Ans:

- In a structure, each member has separate memory, and all members can store values simultaneously.
- In a union, all members share the same memory, so only one member's value should be used at a time.

7. What are the operators used to access structure members?

Ans: Two operators are used:

- Dot operator (`.`) – accesses members using a structure variable.
- Arrow operator (`->`) – accesses members using a structure pointer.

8. What is an array of structures?

Ans: An array of structures stores multiple structure variables of the same type under one array name.

Example: struct Student s[5];

9. List out the applications of structures.

Ans:

1. Employee and student databases.
2. Library management systems.
3. Storing records with different data types.
4. Representing complex data.

10. How can you access a structure member through a pointer?

Ans: We can access a structure member through a pointer in two ways:

1. Using the dereference and dot operators: (*ptr).name
2. Using the arrow operator: ptr->name